Snowflake Feature Catalog Requirements

Context

While this HLDD outlines the future state for the Pilot domain (encompassing Experiment Analysis and Feature Catalog services), migration efforts are only required for the Feature Catalog Service.

Currently all Feature Catalog Snowflake assets are available as dbt models in the feature_catalog v2 namespace. The HLDD details the target state split of this into Conformed and Metrics Layers, including the outputs of many ongoing projects (e.g. Performance & Reliability metrics, Cost metrics).

Below we focus on the immediate requirements within scope for Snowflake's data migration work.

Requirements

Full diagram below on page 5.

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1. Creation of **Feature Catalog Metrics Layer** (highlighted in yellow above, *see callout #1* below).

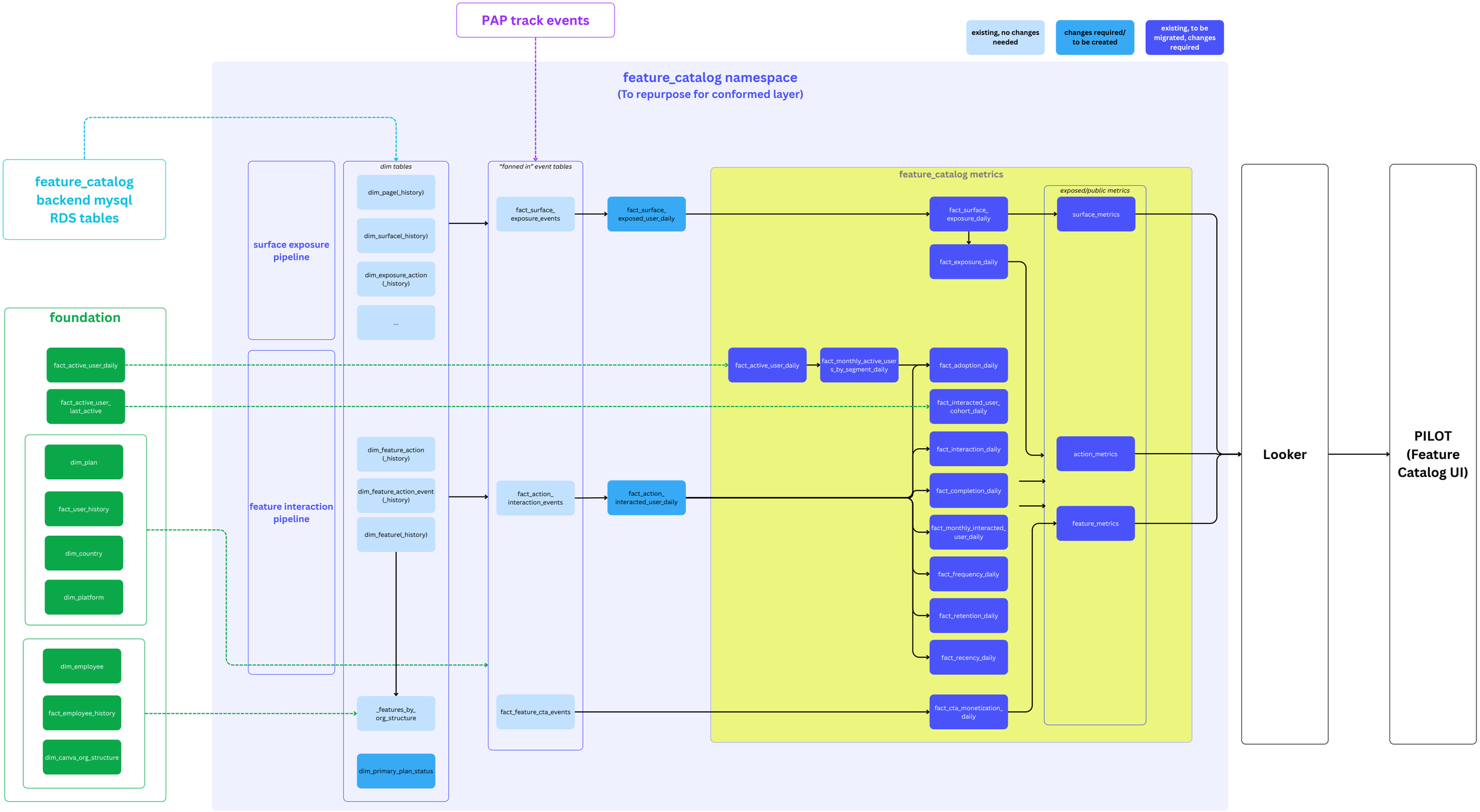
2. **Migration** of specified models from the existing feature_catalog namespace to the Feature Catalog Metrics Layer (16 models, see exact list on page 4).
3. Create **dim_primary_plan_status** in the Feature Catalog conformed layer, for use in #4.
4. Conform the specified models to **dimensional modelling**.
   a. For the specified 18 models: add in foreign/surrogate dkeys from events/foundation dimension tables where possible, i.e. dim_platform_dkey (platform), dim_plan_dkey (primary_plan_type), dim_primary_plan_status_dkey (see #3).
   b. For the specified 3 exposed models: add in foreign dkeys for Feature Catalog entities (i.e. dim_feature_dkey, dim_feature_action_dkey, dim_surface_dkey).
   c. *Note: This will involve both dbt model logic changes and populating new columns for the existing tables.*
5. ~~For the specified 3 exposed models, create **Snowflake semantic views**.~~

Callouts

1. The infrastructure architecture for Conformed and Metrics Layers has yet to be confirmed, i.e. whether they will exist within a single database, or should be split into separate databases.
2. The Pilot Domain HLDD is to be reviewed by the Analytics Engineering Working Group from 20 Feb, so is subject to change - work should not commence until this has been reviewed.
3. Feature Catalog is not a critical service, as such there is no urgent deadline for this work.
4. However, since Feature Catalog metrics are integrated with a backend/frontend service (via Looker) that supports an internal tool, we would like to avoid any downtime of our Looker dashboards in particular during the migration.

Existing FC Model in feature_catalog namespace	Conformed/Metric	Exposed?	Changes required		*Will exist in 1-2 weeks*
dim_feature	conformed	yes	none		
dim_feature_history	conformed	yes	none		
dim_feature	conformed	yes	none		
dim_feature_action	conformed	yes	none		
dim_feature_action_history	conformed	yes	none		
dim_feature_action_event_history	conformed	yes	none		
dim_feature_action_surface_mapping	conformed	yes	none		
dim_surface	conformed	yes	none		
dim_surface_history	conformed	yes	none		
dim_feature_action_surface_mapping	conformed	yes	none		
dim_page	conformed	yes	none		
dim_page_history	conformed	yes	none		
dim_exposure_action	conformed	yes	none		
dim_exposure_action_history	conformed	yes	none		
dim_exposure_action_event	conformed	yes	none		
dim_exposure_action_event_history	conformed	yes	none		
dim_exposure_action_entity	conformed	yes	none		
dim_exposure_action_entity_history	conformed	yes	none		
fact_action_interation_event	conformed	yes	none		
fact_action_interacted_user_daily	conformed	yes	expose surrogate dkeys from events/foundation dimension tabl		
fact_exposure_action_events	conformed	yes	none		
fact_exposure_action_user_segments_daily	conformed	yes	expose surrogate dkeys from events/foundation dimension tabl		
fact_feature_cta_events	conformed	yes	none		

fact_active_user_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_adoption_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_completion_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_cta_monetization_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_frequency_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_interacted_user_cohort_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_interacted_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_monthly_active_users_by_segment_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_monthly_interacted_user_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_recency_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_retention_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_surface_exposure_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
fact_exposure_daily	metric	no	expose surrogate dkeys from events/foundation dimension tabl
surface_metrics	metric	yes	expose surrogate dkeys from events/foundation dimension tabl expose surrogate dkeys for surface dimension create semantic view
action_metrics	metric	yes	expose surrogate dkeys from events/foundation dimension tabl expose surrogate dkeys for surface dimension create semantic view
feature_metrics	metric	yes	expose surrogate dkeys from events/foundation dimension tabl expose surrogate dkeys for surface dimension create semantic view


PAP track events
existing, no changes needed
changes required/ to be created
existing, to be migrated, changes required
feature_catalog namespace
(To repurpose for conformed layer)
feature_catalog backend mysql RDS tables
foundation
fact_active_user_daily
fact_active_user_ last_active
dim_plan
fact_user_history
dim_country
dim_platform
dim_employee
fact_employee_history
dim_canva_org_structure
surface exposure pipeline
feature interaction pipeline
dim tables
dim_page(_history)
dim_surface(_history)
dim_exposure_action (_history)
...
dim_feature_action (_history)
dim_feature_action_event (_history)
dim_feature(_history)
_features_by_ org_structure
dim_primary_plan_status
"fanned in" event tables
fact_surface_ exposure_events
fact_action_ interaction_events
fact_feature_cta_events
fact_surface_ exposed_user_daily
fact_action_ interacted_user_daily
feature_catalog metrics
fact_surface_ exposure_daily
fact_exposure_daily
fact_active_user_daily
fact_monthly_active_user s_by_segment_daily
fact_adoption_daily
fact_interacted_user_ cohort_daily
fact_interaction_daily
fact_completion_daily
fact_monthly_interacted_ user_daily
fact_frequency_daily
fact_retention_daily
fact_recency_daily
fact_cta_monetization_ daily
exposed/public metrics
surface_metrics
action_metrics
feature_metrics
Looker
PILOT (Feature Catalog UI)